

AP CALCULUS AB - UNIT 2

Guided Notes 2.5: Product and Quotient Rules



Name: _____ Class: _____ Date: _____

Learning Objectives and Essential Question

Essential question: How can we differentiate products of differentiable functions?

- Differentiate products of differentiable functions.
- Use values and derivative values from tables or graphs in the Product Rule.
- Choose between expanding first and applying the Product Rule.
- Differentiate quotients of differentiable functions.
- Apply the Quotient Rule to symbolic, tabular, and contextual representations.
- Respect denominator restrictions and simplify strategically.

Key Knowledge, Vocabulary, and Notation

Product Rule: Product Rule

$$(fg)' = f'g + fg'.$$

Product Rule: At a point

$$(fg)'(a) = f'(a)g(a) + f(a)g'(a).$$

Product Rule: Three factors

$$(fgh)' = f'gh + fg'h + fgh'.$$

Product Rule: Common error

In general, $(fg)' \neq f'g'$.

Quotient Rule: Quotient Rule

$$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}, \text{ where } g \neq 0.$$

Quotient Rule: Memory structure

Low times derivative of high minus high times derivative of low, over low squared.

Quotient Rule: At a point

All four values $f(a)$, $f'(a)$, $g(a)$, $g'(a)$ are needed unless a term vanishes.

Quotient Rule: Domain

The original denominator and the squared denominator in the derivative cannot be zero.

Concept Development and Strategy

1. **Product Rule.** Name the factors before differentiating. Keep one factor unchanged while differentiating the other, then reverse roles and add.
2. **Product Rule.** If both factors are simple polynomials, expansion can be faster and provides a useful check. If factors have meaningful structure or include transcendental functions, the Product Rule is usually cleaner.
3. **Product Rule.** Table problems require four values at the target: f , f' , g , and g' .
4. **Quotient Rule.** Write the denominator squared first, then construct the ordered numerator. The minus sign is a frequent source of errors.

5. **Quotient Rule.** Before using the rule, check whether algebraic simplification turns the quotient into a sum of powers. Simplifying can be faster, but domain restrictions from canceled factors must still be remembered.
6. **Quotient Rule.** A horizontal tangent occurs where the derivative numerator is zero while the original denominator is nonzero.

Shared Representation / Data

Product Rule

x	-1	0	1	2
$f(x)$	3	-2	4	5
$f'(x)$	-1	3	2	-4
$g(x)$	2	5	-1	3
$g'(x)$	4	-2	6	1

Quotient Rule

x	-1	0	1	2
$f(x)$	3	-2	4	5
$f'(x)$	-1	3	2	-4
$g(x)$	2	5	-1	3
$g'(x)$	4	-2	6	1

Worked Example: Product Rule - Symbolic product

Differentiate $f(x) = x^3 \sin x$.

Answer and Reasoning

$$f'(x) = 3x^2 \sin x + x^3 \cos x.$$

Worked Example: Product Rule - Table values

If $u(2) = 3$, $u'(2) = -1$, $v(2) = 5$, and $v'(2) = 4$, find $(uv)'(2)$.

Answer and Reasoning

$$(-1)(5) + (3)(4) = 7.$$

Worked Example: Product Rule - Expand or use the rule

Differentiate $(x^2 + 1)(x - 4)$.

Answer and Reasoning

$$\text{Product Rule: } 2x(x - 4) + (x^2 + 1) = 3x^2 - 8x + 1.$$

Worked Example: Quotient Rule - Symbolic quotient

Differentiate $f(x) = \frac{x^2+1}{x-3}$.

Answer and Reasoning

$$f'(x) = \frac{2x(x-3)-(x^2+1)}{(x-3)^2} = \frac{x^2-6x-1}{(x-3)^2}.$$

Worked Example: Quotient Rule - Table values

If $u(1) = 4$, $u'(1) = 2$, $v(1) = -1$, and $v'(1) = 3$, find $(u/v)'(1)$.

Answer and Reasoning

$$\frac{2(-1)-4(3)}{(-1)^2} = -14.$$

Worked Example: Quotient Rule - Simplify first

Differentiate $\frac{x^4-2x}{x^2}$ for $x \neq 0$.

Answer and Reasoning

Rewrite as $x^2 - 2x^{-1}$, so the derivative is $2x + 2x^{-2}$.

Guided Practice and Discussion

Directions

Show the mathematical setup, preserve exact values unless a decimal is requested, and justify existence or interpretation statements with the appropriate definition or theorem.

1. Product Rule: Differentiate x^2e^x .

2. Product Rule: Differentiate $(x^3 + 1) \sin x$.

3. Product Rule: Differentiate $f(x) = x^4 \ln x$ for $x > 0$.

4. Product Rule: Find the tangent slope of $y = (x^2 + 2)e^x$ at $x = 0$.

5. Product Rule: Find $(f^2)'(a)$ in terms of $f(a)$ and $f'(a)$ without invoking the Chain Rule.

6. Product Rule: Given $F = fg$, $F(a) = 0$, and $F'(a) \neq 0$, what can be concluded about $f(a)$, $g(a)$?

7. Product Rule: Prove the Product Rule from the limit definition by adding and subtracting $f(x+h)g(x)$.

AP Reasoning Check

Multiple-Choice Check

Choose the best answer. Explain why at least one distractor is incorrect during class discussion.

1. Product Rule: $(fg)' =$

(A) $f'g'$

(B) $f'g + fg'$

(C) $fg' + g$

(D) $f' + g'$

2. Product Rule: If $f(0) = 2$, $f'(0) = 3$, $g(0) = 4$, $g'(0) = -1$, then $(fg)'(0) =$

(A) 5

(B) 10

(C) 12

(D) 14

3. Product Rule: Using the shared table, $(fg)'(2)$ equals

(A) -17

(B) -7

(C) 17

(D) 23

Common Errors and AP Precision

- State the derivative notation, input value, sign, and units when interpreting a rate in context.
- Check differentiability before using a tangent slope, and keep the order of Product and Quotient Rule terms accurate.
- A complete AP justification names the relevant definition, condition, or theorem rather than reporting only a numerical result.

Exit Ticket

Quotient Rule: Find a condition for f/g to have a stationary point at a in terms of relative rates, assuming $f(a), g(a) \neq 0$.

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